

Foundation (Jalapeno)

Q1. Simplify $\sqrt{9} + \sqrt{16}$

- (A) $3 + 4$
- (B) 5
- (C) 7
- (D) $\sqrt{25}$
- (E) $1 + 2 \cdot \sqrt{2}$

Q2. Find $\sqrt{4} \cdot \sqrt{9}$

- (A) $\sqrt{36}$
- (B) $2 \cdot \sqrt{3}$
- (C) 6
- (D) $\sqrt{12}$
- (E) 3

Q3. A space lab measures the distance to a planet as $\sqrt{100} + \sqrt{25}$ kilometers. Simplify the expression.

- (A) 15 km
- (B) $10 + 5$ km
- (C) $5 \cdot \sqrt{5}$ km
- (D) $\sqrt{125}$ km
- (E) 20 km

Q4. A physics experiment involves a force of $\sqrt{16} - \sqrt{9}$ Newtons. Simplify the expression.

- (A) 1 N
- (B) 5 N
- (C) $\sqrt{7}$ N
- (D) -1 N
- (E) $4 - 3$ N

Q5. Simplify $\frac{\sqrt{36}}{\sqrt{4}}$

- (A) $\sqrt{9}$
- (B) 3
- (C) 6
- (D) $\frac{6}{2}$
- (E) $\frac{\sqrt{36}}{2}$

Q6. A spacecraft travels $\sqrt{49} + \sqrt{25}$ kilometers in a day. Simplify the expression.

- (A) 12 km
- (B) $7 + 5$ km
- (C) $\sqrt{74}$ km
- (D) $\sqrt{50}$ km
- (E) 10 km

Q7. Find $\sqrt{64} \cdot \sqrt{9}$

- (A) 24
- (B) $\sqrt{576}$
- (C) $8 \cdot \sqrt{9}$
- (D) $6 \cdot \sqrt{16}$
- (E) 12

Q8. A lab measures the energy of a reaction as $\sqrt{25} - \sqrt{16}$. Simplify the expression.

- (A) 3
- (B) 2
- (C) 1
- (D) $\sqrt{9}$
- (E) $\sqrt{25} - 4$

Open Ended Questions (FRQ)

Q9. Simplify $\frac{\sqrt{128}}{\sqrt{8}}$. Show your work and explain your reasoning.

Q10. A space station has a rectangular solar panel with an area of $\sqrt{256}$ square meters. If the length of the panel is $\sqrt{16}$ meters, find the width of the panel. Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Remind students to simplify radicals before adding or subtracting.
- Tip 2: Encourage students to use the properties of radicals to simplify expressions.
- Tip 3: Emphasize the importance of showing work and explaining reasoning in free-response questions.